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From Raw Sales to Smart Insights: KPIs, DAX Calculations and an Interactive Power BI Dashboard on the Adventure Works Dataset

Symbiosis Institute of Technology, Pune | Department of AI & ML | Sem V | AY 2026-27 |

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1. Introduction

Raw sales data does not tell a story by itself. A business needs numbers like total revenue, profit, order count and growth rate before it can take decisions. In this practical, I used Power BI Desktop and DAX Studio to turn the Adventure Works sales data into measurable KPIs, wrote custom DAX measures, tested them with DAX queries, and finally built an interactive dashboard that anyone can filter by year, product category and region.

2. Objective

- Create KPI measures: Total Sales, Total Profit, Total Orders, Total Customers, Average Sales, Average Order Value and Profit Margin (%).
- Write more than 20 custom DAX measures using SUM, AVERAGE, COUNT, DISTINCTCOUNT, CALCULATE, DIVIDE, FILTER and RANKX.
- Perform time-based analysis: monthly sales, previous-period sales, sales variance and growth %.
- Validate the measures and check query performance in DAX Studio.
- Build an interactive dashboard and interpret the results.

3. Dataset and Tools

Dataset: Adventure Works (bicycle company sales data, 2020 to 2022).

Tables used: Sales_2020_2022 (fact table), Products, Product_Subcategories, Product_Categories, Customers, Territories, Calendar and Returns.

Tools: Power BI Desktop (data model, measures and dashboard) and DAX Studio 3.6.1 (query execution, validation and performance analysis).

4. Steps Performed

Step 1: Load data and build the data model

I imported the Adventure Works tables into Power BI Desktop and created relationships so that filters flow from the dimension tables to the sales table: Products to Product_Subcategories to Product_Categories, Customers to Sales, Territories to Sales, and Calendar to Sales (on OrderDate). The Calendar table was marked as the Date table so that time-intelligence functions work correctly.

Step 2: Create the 10 KPI measures

These are the measures required by the assignment. Sales is calculated as quantity multiplied by product price, and cost as quantity multiplied by product cost. DIVIDE() is used instead of the "/" operator so a division by zero returns blank rather than an error.

No.	Measure	DAX formula
1	**Total Sales**	`SUMX(Sales_2020_2022, Sales_2020_2022[OrderQuantity] * RELATED(Products[ProductPrice]))`
2	**Total Cost**	`SUMX(Sales_2020_2022, Sales_2020_2022[OrderQuantity] * RELATED(Products[ProductCost]))`
3	**Total Profit**	`[Total Sales] - [Total Cost]`
4	**Total Quantity**	`SUM(Sales_2020_2022[OrderQuantity])`
5	**Total Orders**	`DISTINCTCOUNT(Sales_2020_2022[OrderNumber])`
6	**Total Customers**	`DISTINCTCOUNT(Sales_2020_2022[CustomerKey])`
7	**Average Sales**	`AVERAGEX(Sales_2020_2022, Sales_2020_2022[OrderQuantity] * RELATED(Products[ProductPrice]))`
8	**Average Order Value**	`DIVIDE([Total Sales], [Total Orders])`
9	**Average Product Price**	`AVERAGE(Products[ProductPrice])`
10	**Profit Margin %**	`DIVIDE([Total Profit], [Total Sales]) * 100`

Step 3: Create additional custom DAX measures

To go beyond the basics, I wrote 12 more measures covering counts, averages, returns, filtered calculations (CALCULATE with FILTER), percentage contribution and ranking (RANKX).

No.	Measure	DAX formula
11	**Total Order Lines**	`COUNTROWS(Sales_2020_2022)`
12	**Average Product Cost**	`AVERAGE(Products[ProductCost])`
13	**Avg Quantity per Order**	`DIVIDE([Total Quantity], [Total Orders])`
14	**Sales per Customer**	`DIVIDE([Total Sales], [Total Customers])`
15	**Total Returns**	`SUM(Returns[ReturnQuantity])`
16	**Return Rate %**	`DIVIDE([Total Returns], [Total Quantity]) * 100`
17	**Bikes Sales**	`CALCULATE([Total Sales], Product_Categories[CategoryName] = "Bikes")`

No.	Measure	DAX formula
18	**High Value Sales**	`CALCULATE([Total Sales], FILTER(Products, Products[ProductPrice] > 1000))`
19	**Sales Contribution %**	`DIVIDE([Total Sales], CALCULATE([Total Sales], ALLSELECTED())) * 100`
20	**Product Sales Rank**	`RANKX(ALL(Products[ProductName]), [Total Sales], , DESC, Dense)`
21	**Category Sales Rank**	`RANKX(ALL(Product_Categories[CategoryName]), [Total Sales], , DESC, Dense)`
22	**Region Sales Rank**	`RANKX(ALL(Territories[Region]), [Total Sales], , DESC, Dense)`

Step 4: Time-based analysis

Using the Calendar table, I created measures for monthly sales, previous-period sales, variance and growth percentage. Variance shows the absolute change, and growth % shows how big that change is relative to the previous period.

No.	Measure	DAX formula
23	**Monthly Sales**	`TOTALMTD([Total Sales], 'Calendar'[Date])`
24	**Previous Month Sales**	`CALCULATE([Total Sales], DATEADD('Calendar'[Date], -1, MONTH))`
25	**Sales Variance**	`[Total Sales] - [Previous Month Sales]`
26	**Sales Growth %**	`DIVIDE([Sales Variance], [Previous Month Sales]) * 100`
27	**YTD Sales**	`TOTALYTD([Total Sales], 'Calendar'[Date])`
28	**Previous Year Sales**	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar'[Date]))`
29	**YoY Growth %**	`DIVIDE([Total Sales] - [Previous Year Sales], [Previous Year Sales]) * 100`

Total measures created: 29 (10 KPI measures + 12 additional measures + 7 time-intelligence measures), which is more than the 20 required.

Step 5: Validate measures and analyse performance in DAX Studio

I connected DAX Studio to the Power BI model (localhost) and ran DAX queries to check that the measures return correct values outside the report. The first query returns the Top 10 products by sales using TOPN and SUMMARIZECOLUMNS. The Server Timings trace was turned on to study how the query executes.

```
EVALUATE
TOPN(
    10,
    SUMMARIZECOLUMNS(
        Products[ProductName],
        "Sales", [Total Sales],
        "Profit", [Total Profit]
    ),
    [Sales], DESC
)
```

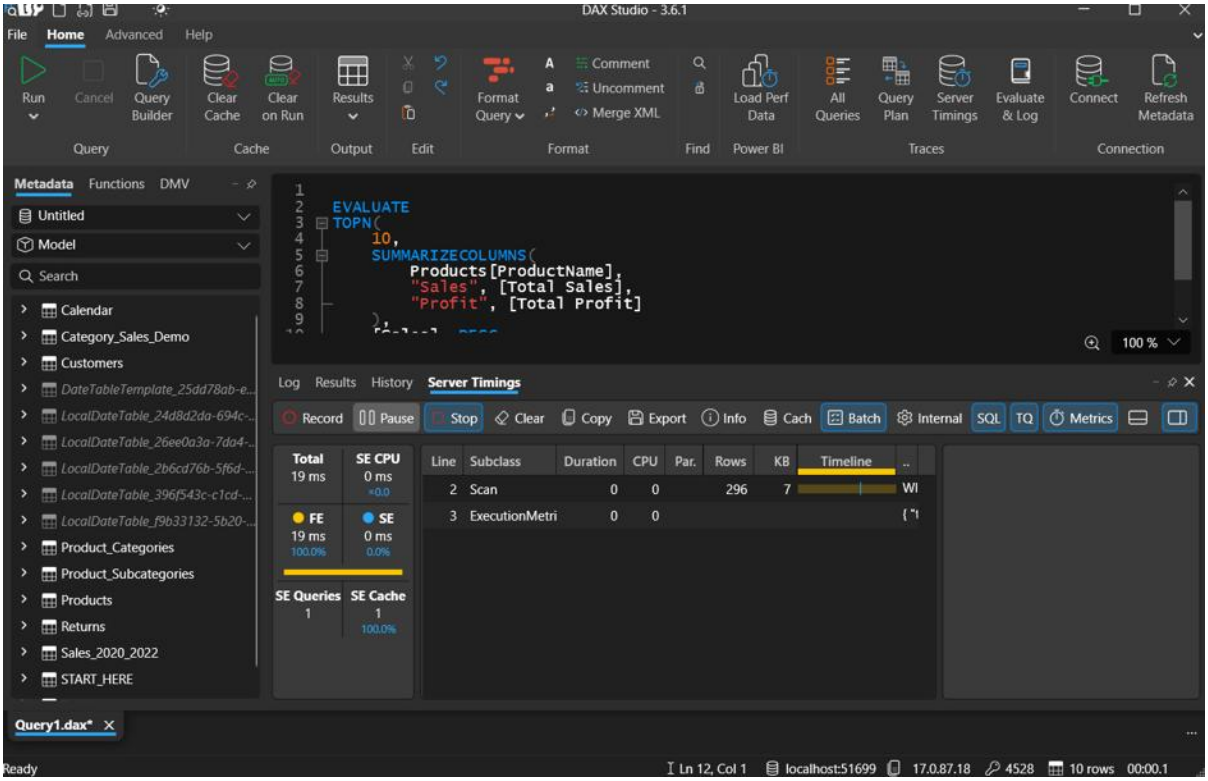


Figure 1: DAX Studio – Top 10 products query with Server Timings

Performance reading: The query finished in just 19 ms in total. The Formula Engine (FE) took all 19 ms (100%) and the Storage Engine (SE) took 0 ms, with only 1 SE query that was served from the SE cache (100%). The scan returned 296 rows (7 KB). This shows the measures are efficient and the model is well optimised for this query.

The second query validates the measures at category level, including the Profit Margin %. The results were sorted by Sales in descending order.

```
EVALUATE
SUMMARIZECOLUMNS(
    Product_Categories[CategoryName],
    "Sales", [Total Sales],
    "Profit", [Total Profit],
    "Profit Margin %", [Profit Margin %]
)
ORDER BY [Sales] DESC
```

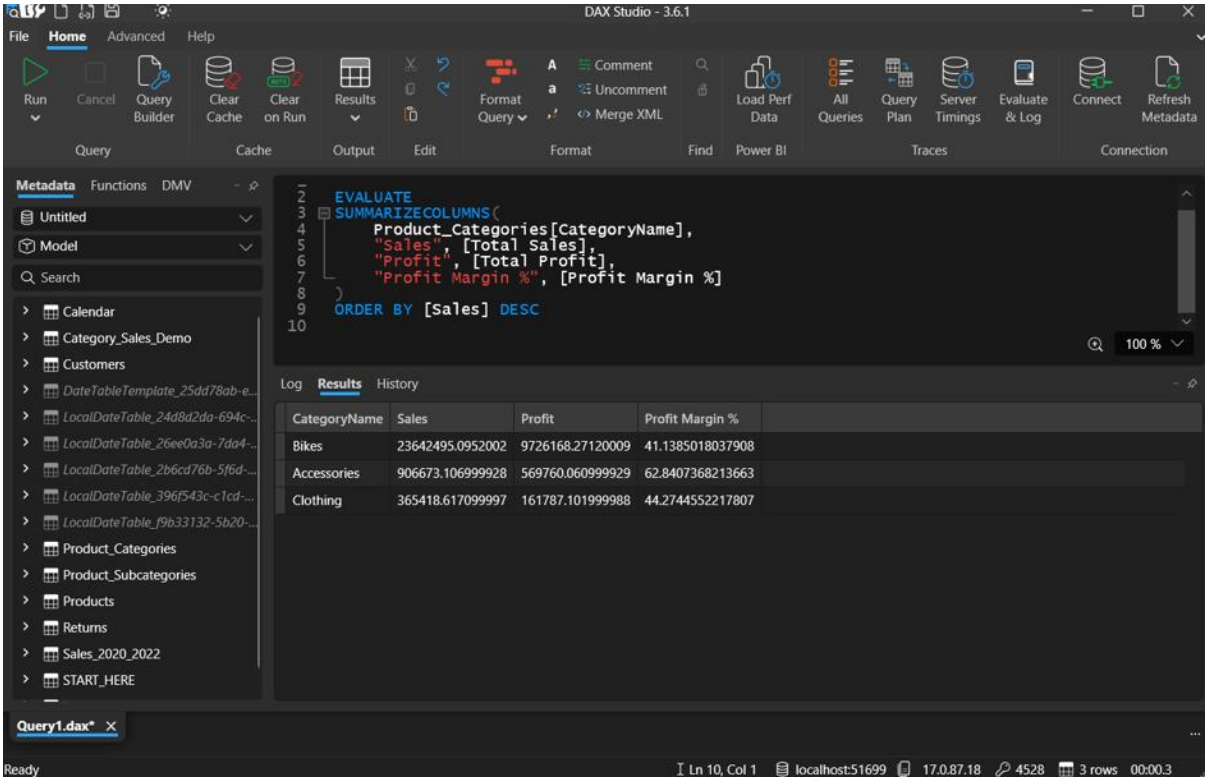
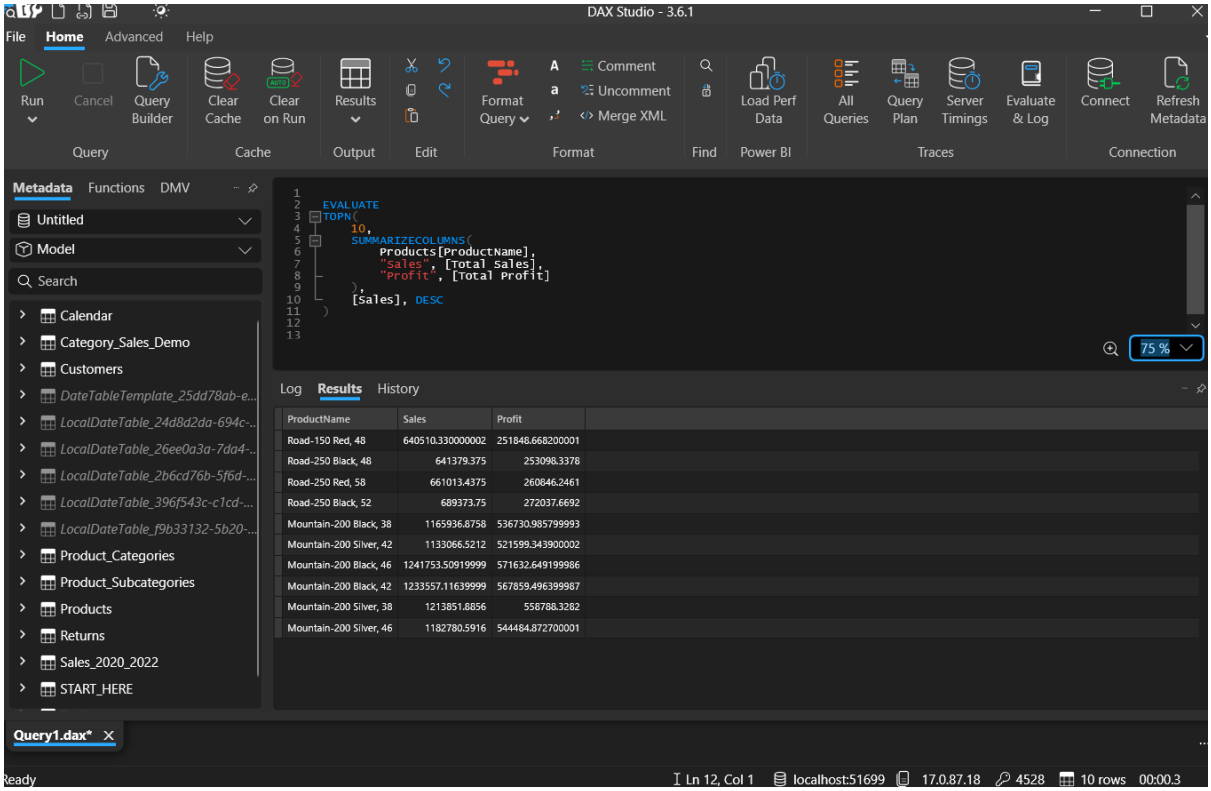


Figure 2: DAX Studio – Category-wise Sales, Profit and Profit Margin %



Top 10 Products by Sales and Profit

Category	Sales	Profit	Profit Margin %
Bikes	23,642,495	9,726,168	41.14%
Accessories	906,673	569,760	62.84%
Clothing	365,418	161,787	44.27%

The category totals add up to the overall figures shown on the dashboard (Sales about 24.91M and Profit about 10.46M), which confirms that the measures are correct.

Step 6: Build the interactive dashboard

In Power BI Desktop, I added KPI cards for the main measures, followed by four visuals and three slicers:

- **KPI cards:** Total Sales, Profit Margin %, Total Customers, Total Orders, Total Profit and Average Order Value.
- **Bar chart:** Total Sales by ProductName (top products).
- **Line chart:** Total Sales by OrderDate (sales trend).
- **Bar chart:** Total Sales by CategoryName.
- **Column chart:** Total Sales by Region.
- **Slicers:** Region, CategoryName and Year. Selecting any value updates all cards and charts together.

5. Output: Dashboard with KPI Cards

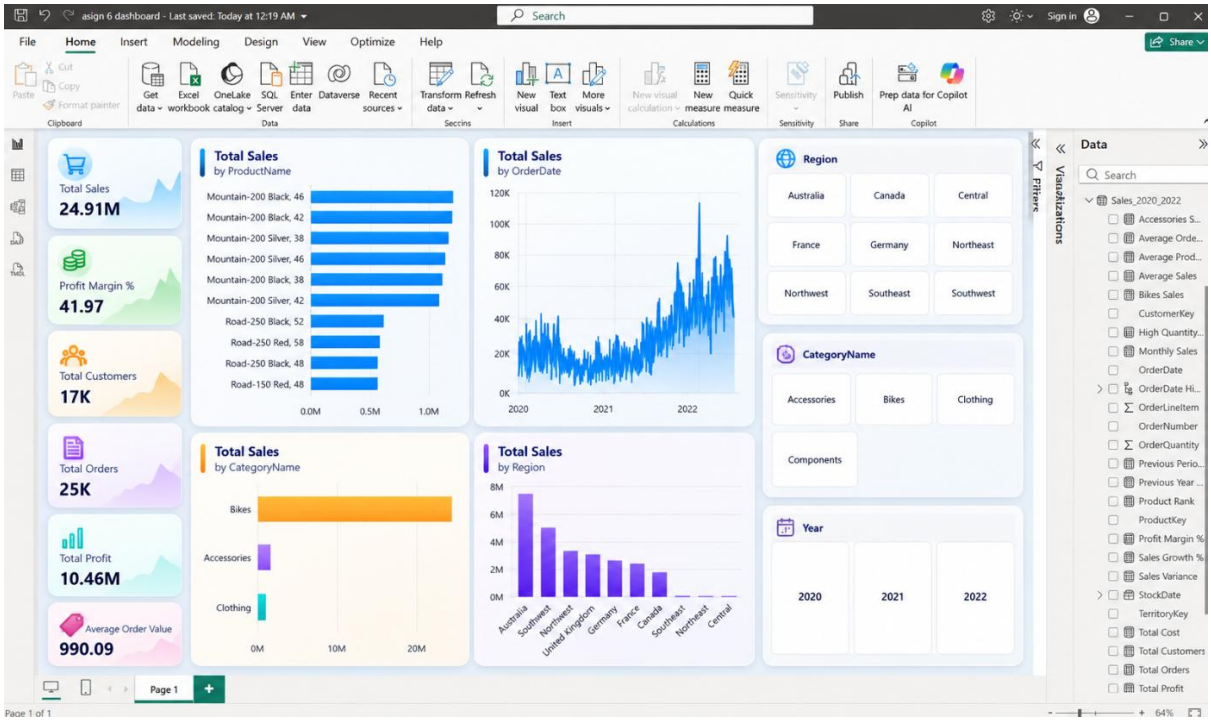


Figure 3: Interactive Power BI dashboard with KPI cards, charts and slicers

KPI	Value	Meaning
Total Sales	24.91M	Overall revenue generated 2020–2022
Total Profit	10.46M	Sales minus product cost
Profit Margin %	41.97	Healthy margin across the business
Total Orders	25K	Distinct order numbers
Total Customers	17K	Unique customers
Average Order Value	990.09	Revenue earned per order

6. Analysis and Interpretation

Top-selling products

The Mountain-200 series dominates the sales chart. Mountain-200 Black (sizes 46 and 42) and Mountain-200 Silver (sizes 38 and 46) are at the top with sales of roughly 1.0M to 1.2M each. The Road-250 and Road-150 models follow far behind at around 0.6M to 0.7M. A few premium mountain bikes therefore bring in a large part of the revenue.

Highest-performing categories

Bikes generate about 23.64M of the 24.91M total sales, which is around 95% of all revenue. Accessories (0.91M) and Clothing (0.37M) are very small in comparison. In terms of profitability, however, Accessories have the best margin at 62.84%, compared to 44.27% for Clothing and 41.14% for Bikes. Bikes earn the most profit in absolute terms (9.73M), while Accessories are the most profitable per rupee/dollar sold.

Highest-performing regions

Australia is the top region with about 7.4M in sales, followed by Southwest (about 4.8M), Northwest (about 3.1M), United Kingdom (about 2.9M), Germany (about 2.5M), France (about 2.4M) and Canada (about 1.8M). The Southeast, Northeast and Central regions contribute almost nothing.

Sales trends

Daily sales stayed low and fairly flat through 2020 and the first half of 2021, mostly in the range of about 5K to 30K. From mid-2021 sales began to rise sharply and reached peaks above 90K to 110K per day in 2022. The business was clearly growing, with high day-to-day fluctuation.

Profitability

The overall profit margin is 41.97% on sales of 24.91M, giving a total profit of 10.46M. The average order value is 990.09 across about 25K orders from 17K customers, which means the average order is high-value because bicycles are expensive items.

7. Key Insights

- Bikes are the core of the business (about 95% of sales), so any change in bike demand strongly affects total revenue.
- Accessories have the highest margin (62.84%) but a very low sales volume, so there is a good opportunity to grow them by bundling with bike purchases.
- Sales are concentrated in a few regions, mainly Australia and the US Southwest, while some regions contribute almost nothing and need market development or a data check.
- Sales grew strongly from mid-2021 onwards, showing positive momentum for 2022.
- Revenue is concentrated in a handful of Mountain-200 products, which is a risk if their demand drops.

8. Conclusion

In this practical, I created 29 DAX measures (10 required KPIs, 12 additional measures and 7 time-intelligence measures) and validated them in DAX Studio with queries and server timings. I then built an interactive Power BI dashboard with KPI cards, four visualisations and slicers for year, category and region. The analysis shows that Adventure Works is a growing, bike-driven business with a healthy 41.97% profit margin, strong results in Australia and the US, and a clear opportunity to increase profit through high-margin accessories and expansion in weaker regions.